GSoC'26 Project Proposal

SugarLabs: Sugarizer ConnectTheDots

Rejah Rabeeul Haque

Basic Details

Full Name:

Rejah Rabeeul Haque

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Your First Language:

My first language is Malayalam but I am proficient in speaking, reading, writing, and understanding English.

Location and Timezone:

Location: Kozhikode, Kerala, India
Timezone: Indian Standard Time (UTC+5:30)

Communication:

The time I will be comfortable working with:

- UTC 22:30 – 02:00 (IST 04:00 – 07:30)
- UTC 08:00 – 12:00 (IST 13:30 – 17:30)
- UTC 16:30 – 19:00 (IST 22:00 – 00:30)

I can be flexible on my day 1 hours early or late if it helps to communicate with other developers and mentors, and I will be reachable anytime through my Mobile No, Discord and Email.

Education Details

I am currently in my third year of pursuing a Bachelor of Technology in Computer Science and Engineering from KMCT Institute of Emerging Technologies and Management under **APJ Abdul Kalam Technological University**. I have started programming in my second year, and I have gained a lot of experience building up my skills in Data Structures and Algorithms, Operating Systems, Cyber Security, and Web Development. I have also been able to apply my skills through various hackathons and projects within my home state throughout the years. Lately, I have focused on learning more about AI and Machine Learning currently, I developed an algorithm for predicting Road Accidents using historical records, weather, and road characteristics and time factors that contribute to where accidents may occur. I have begun to contribute to Open Source projects since January 2026, specifically with Sugar Labs (Sugarizer). In this project, I am developing enhancements for web-based tools for education. I have experience programming mainly in C/C++, Python, and new web application development, I am also an eSports professional, which has helped me build my teamwork capabilities outside of school.

Share links, if any, of your previous work on open source projects

I started my open source journey with SugarLabs in my third year of undergraduate studies. After that, I learned more about the community, its structure, and the way people interact in this field.

Since then, I have been continuously contributing to Sugarizer by SugarLabs for nearly 3 months. During this time, I have contributed to many activities such as Fototoon, Chess, Paint, Flip, Calculate, and FractionBounce — fixing bugs, adding enhancements, and implementing new features. I have also started working on my own activity, ColorLab (which is nearly complete), that helps kids learn about color theory. This has also given me a deeper understanding of how Sugarizer works internally.

Sugar Labs:

Pull request link	Description	Status
#1984	Created a activity ColorLab that helps kids to learn about color theory	Pending Merge

#1916	Added Resize feature on Fototoon activity	Merged
#1954	Bug in Fototoon Window resize	Merged
#2021	Enhanced Features on Fototoon activity Globes not displayed when exporting a single image in fototoon	Merged
#1948	Fix: In calculate activity Delete key deletes wrong character issue	Merged
#2041	Enhanced features on Chess activity Saved Chess game fails to restore difficulty setting	Merged
#2058	Fototoon "Clean All" leaves image data in storage while clearing the UI	Merged
#1966	Bug in trigonometric evaluation on Calculate activity	Merged
#2065	Fullscreen Reset Bug in FractionBounce Causes Ball Selection Loss & Bounce Glitch	Pending Merge

I have made over **27 commits** till now and will contribute more.
The most significant contribution was creating a activity ColorLab that helps kids to learn about color theory by interactive games. For this contribution I've added 16,511 lines.

I worked on some of my personal projects for learning purposes and I have also participated in National Cybersecurity Conclave and other hackthons.

Convince us that you will be a good fit for this project, by sharing links to your contribution to Sugar Labs

I am an active contributor of **Sugarizer**(Sugar Labs) since **Jan 2026** with over **27 commits** merged in various activities. Spending this much time with the codebase helped me to understand it in a better way.

- Commits: 27 (16,511++, 321--)
- Issues: 10 (7 closed, 3 open)
- Pull Request: (7 closed, 3 open)

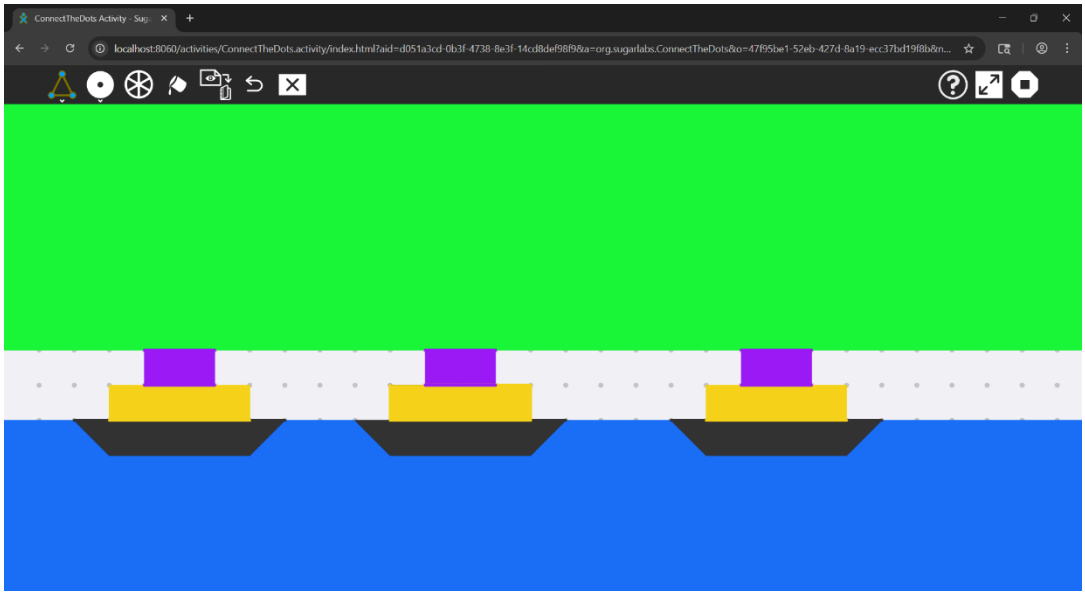
I am applying to Sugarizer (SugarLabs) for Google Summer of Code 2026 because of its strong community relations and the meaningful work carried out by this organization. Here, everyone discusses and develops as a team, and every opinion matters. It truly makes you feel at your best knowing that your work contributes to a social cause and creates a positive impact on society.

Project Details

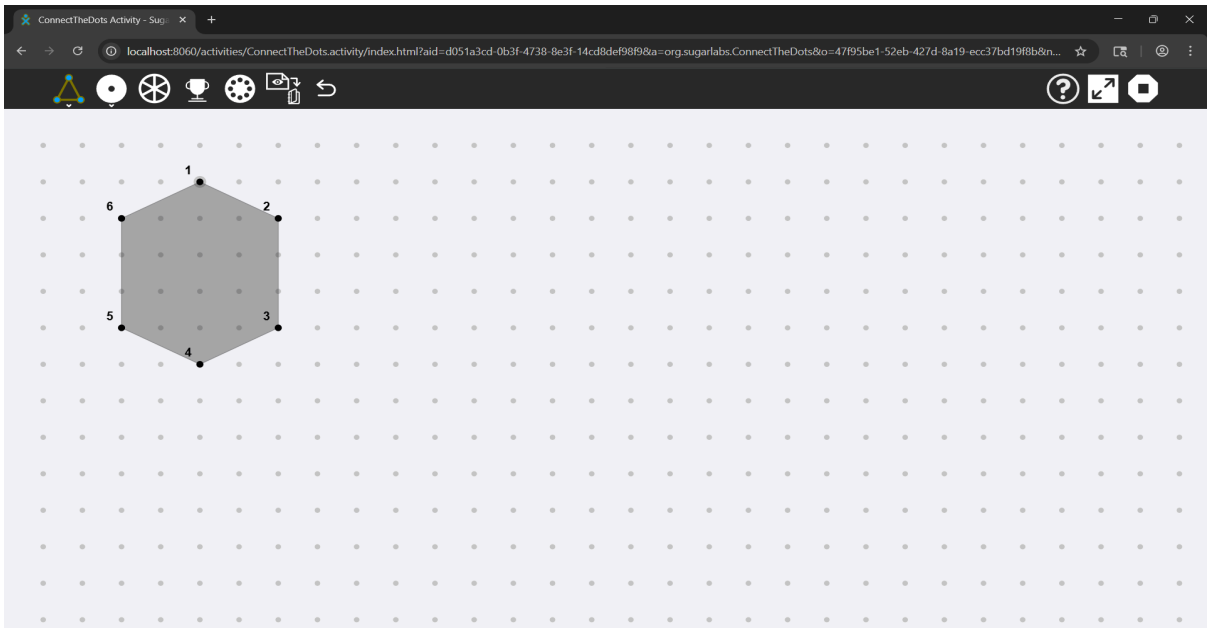
During GSoC, I'll work on Connect the dots activity inspired by the classic "Connect the Dots" game. This activity will combine several modes of interaction to engage users in creative drawing, learning shapes, and strategic gameplay.

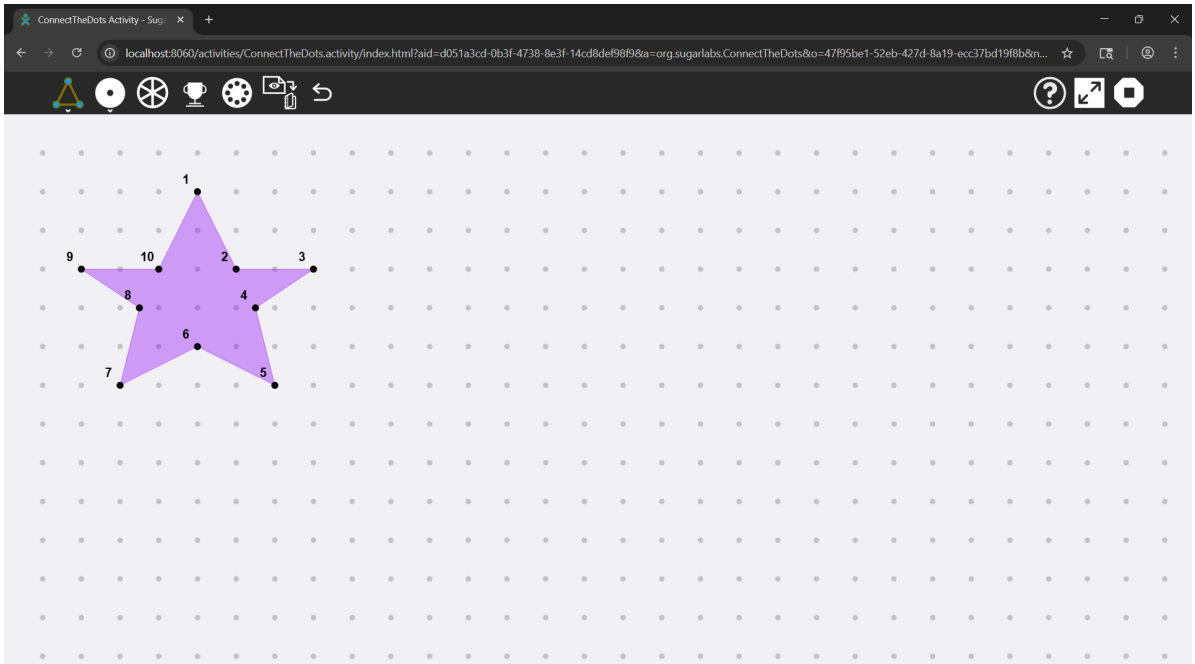
Three distinct modes will be implemented:

- ***Draw mode***: A very basic mode for young users. Let users create freely like in Gridpaint activity - inspired by pok pok. In shared mode, all users are drawing together on the same board like in Paint activity . It will include all necessary features including save their work to the Journal.

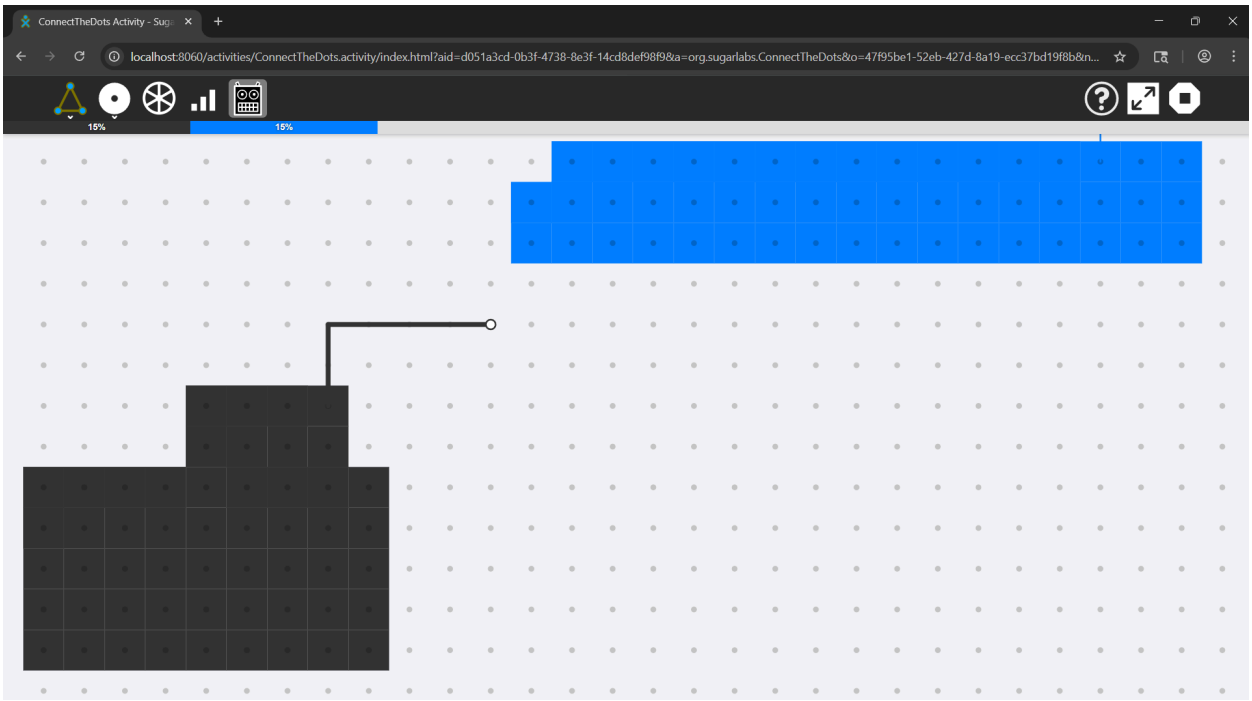


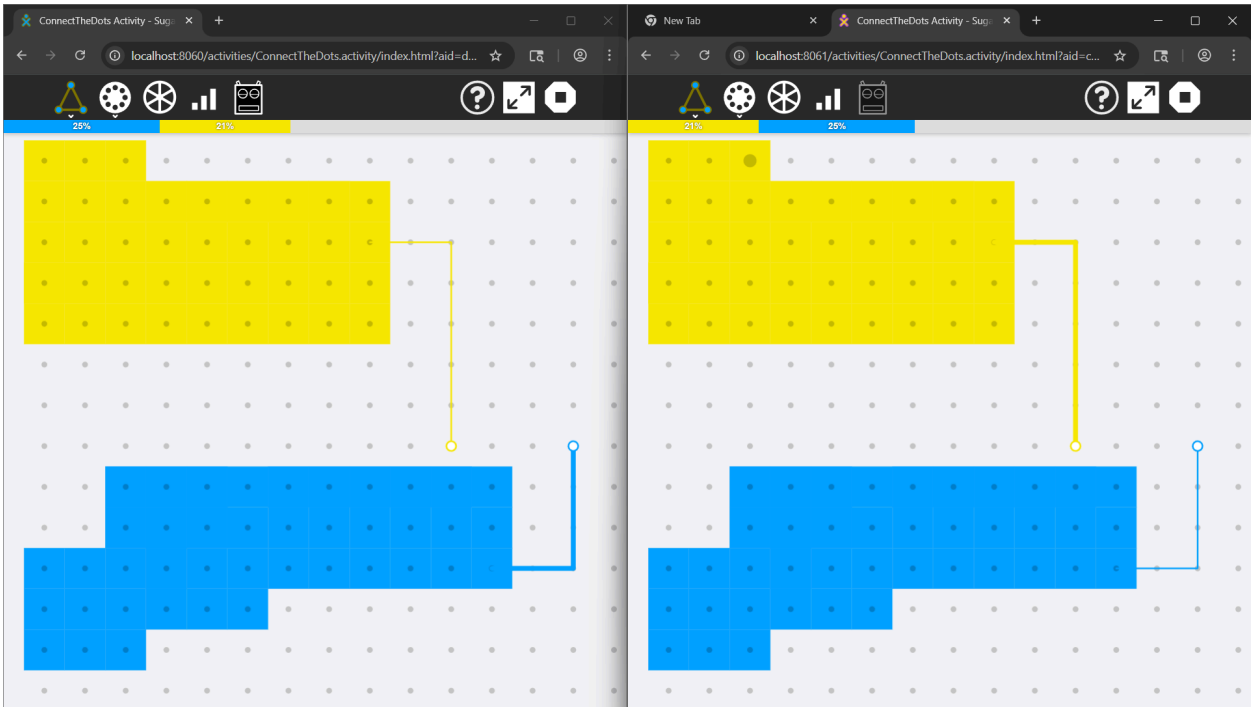
- ***Number mode***: The user chooses a template with numbered dots and has to connect them in the correct order - inspired by connect-the-dots-activity. In shared mode, all users compete to finish the template first like in Maze activity. It should also be possible to create and share new templates like in Calligra or Tangram activities.



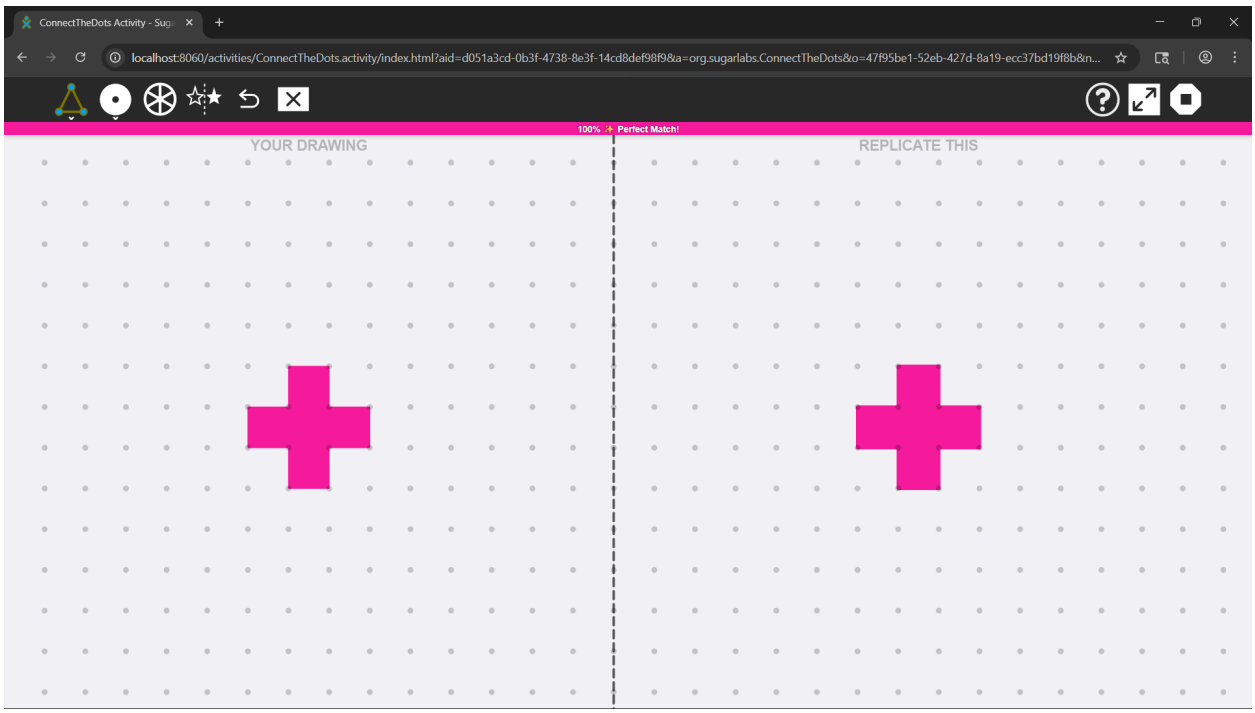


- ***Game mode***: Strategic mode to conquer territory, each users should fill a maximum part of the screen starting from a little square in a border - inspired by [paper io](). If the user is alone, an AI opponent will try to beat him. In shared mode, all users compete to conquer the most territory. Also we can enable on disable ai opponent on both modes.





- ***Replicate Mode:*** A simple painting is shown on the right side, and users try to recreate it on the left side of the canvas — inspired by drawinggame.ai. A starting point is provided to guide them, along with a progress bar to track their progress. Since adding 4 modes in a single activity might be complex, we can combine this with the Number Mode by showing the reference drawing on the right side and displaying numbered dots on the left side of the canvas. This mode can be reconsidered if needed, but it could be helpful for beginners to practice drawing using simple references.



How will it impact Sugar Labs?

Sugarizer is a great platform with many features. It helps many students across country to learn things by playing games also ensuring they are not getting addicted by ensuring minimal UI designs. Currently, Sugarizer has so many activities that can help kids to learn many fields by including Connect the Dots activity the biggest differentiator is that no other Sugarizer activity currently covers all three of creativity, structured learning, and strategic gameplay. This ensures that kids aged 5-11 can feel enthusiasm. The AI opponent in Game mode is also entirely new for Sugarizer most shared activities are purely human-vs-human. Adding a bot means a child can practice strategy even without network peers.

What technologies (programming languages, etc.) will you be using?

ConnectTheDots is a simple web application written entirely on Vanilla JS, HTML5, and CSS3, with its core engine relying completely on the HTML5 Canvas 2D API.
ConnectTheDots will completely built by integrate with the Sugar-Web framework, making use of its built in modules. Such as sugar-web/datastore for saving user progress to the Journal, sugar-web/env for matching the user's Sugarizer theme and language, Real-time multiplayer functionality by the Sugarizer Presence API.
On the technical side, ConnectTheDots includes several algorithms such as, Breadth First Search (BFS) for shape detection, Ray Casting for territory calculation, and Greedy Heuristics to drive the AI opponent. The application also integrates supporting libraries such as i18next for multilingual localization, with support for English, French, and Spanish.

Timeline

Break down the entire project into chunks and tell us what will you work on each week.

Days	Tasks
Pre GSoC Current date to May 1, 2026	➢ Completing the ColorLab activity. ➢ Stay connected with the community and contribute to Sugarizer.
Community Bonding Period May 1, 2026 to May 24, 2026	➢ Discuss the creation of the ConnectTheDots activity and learn about it ➢ Discuss about the modes that are going to be in the activity ➢ Also, discuss about how each mode should behave on the shared mode ➢ Explore the tech stack required for the project.
Week 1 May 25, 2026 to May 31, 2026	➢ Create the complete production folder and file structure with all modules separated. ➢ Build the production Canvas engine as a pure rendering module. ➢ Implement responsive dot grid rendering that correctly recalculates dot positions and spacing on window size. ➢ Implement a dirty region render loop so only changed portions of the Canvas repaint each frame.

Week 2 Jun 1, 2026 to Jun 7, 2026	➢ Build the unified input abstraction layer handling mouse and touch events identically. ➢ Implement dot hit detection with a comfortable tolerance radius. ➢ Add hover over the dots and a ghost line that follows the cursor between dots before commits a connection. ➢ Implement line rendering with clean antialiasing and consistent stroke width across all screen densities. ➢ Develop Draw Mode as the first fully playable mode.
Week 3 Jun 8, 2026 to Jun 14, 2026	➢ Implement Number Mode as the second game mode, built using the same Canvas engine and input system. ➢ Number labels are rendered on each dot the user must connect them strictly in ascending numerical order to see the picture also out-of-order connections are rejected ➢ Render the completed drawing progressively as correct connections are made, revealing the hidden picture line by line ➢ Track completion state and display a congratulations confetti animation screen showing time taken when all dots are connected correctly ➢ Develop default set of built-in templates
Week 4 Jun 15, 2026 to Jun 21, 2026	➢ Build the template editor : a canvas where users can place dots freely, assign numbers to define the correct sequence, preview the resulting drawing, and save it as a custom template ➢ Implement template save to sugar-web/datastore as a Journal entry ➢ Build the template gallery UI : a browsable view for all default templates and saved templates in the Journal that users can select and play ➢ Implement template sharing via the Presence API so users can push their custom templates to all peers in a shared session by mirroring how Calligra and Tangram handle shared content ➢ Validate the functionality
Week 5 Jun 22, 2026 to Jun 28, 2026	➢ Implement the paper.io inspired territory expansion mechanic each player starts from a small square and expands their territory by drawing a path that forms a closed loop back to their territory. ➢ When a loop is completed, run BFS(Breadth First Search) flood fill to capture all cells enclosed within the player's territory. ➢ Use Ray Casting to accurately determine which cells are inside the completed loop for complex paths. ➢ Handle the edge case scenario where a player's path crosses their own trail , another user trail or ai's trail

Week 6 Jun 29, 2026 to Jul 4, 2026	➢ Build the Greedy Heuristic AI that evaluates all possible loop completions, scores each by territory area gained, and executes the best option, introduces easy difficulty to feel beatable ➢ Integrate the AI automatically when a player starts Game Mode alone ➢ Implement the scoring system display as a live progress bar for total grid owned by each player ➢ Integrate sugar-web/activity lifecycle and sugar-web/datastore to save and restore the complete game state for all three modes including Draw Mode canvases, Number Mode progress, and Game Mode territory maps ➢ Build the Sugar toolbar with Reset, Mode Switcher, Network, Color and a Robot icon for turning on and off AI also common icons such as full screen, tutorial, stop, activity icon
Phase 1 Evaluation Jul 5, 2026 to Jul 10, 2026	➢ Midterm deliverable: All three modes fully playable with Sugar-Web integration complete.
Week 7 Jul 5, 2026 to Jul 11, 2026	➢ Integrate the Sugarizer Presence API with WebSocket-based session management ➢ Draw Mode multiplayer: all connected players draw on a shared canvas simultaneously every line placed by any player is broadcast and rendered on all peer’s screens in real time ➢ Number Mode multiplayer: All connected players compete to complete the same template simultaneously the first player to correctly connect all dots wins, also custom templates can be shared ➢ Game Mode multiplayer: Turn based territory conquest with all player positions and territory states synced. Also includes there will be an AI toggle for AI opponent ➢ Each player gets their own unique colour for territory players can change their colour but cannot select a colour already chosen by another player.

Week 8 Jul 12, 2026 to Jul 18, 2026	➢ Test all three multiplayer modes on a live Sugarizer server with real devices ➢ Fix state difference: Ensuring that the host's state is authoritative in all conflict scenarios ➢ Handle disconnection in each mode: Draw Mode pauses collaborative drawing, Number Mode continues with rest of the players, Game Mode also continues with rest of the players ➢ On rejoin, immediately push the full current state to the reconnecting player so they are back in sync ➢ Edge case testing: three or more players in all modes, host leaving mid game, slow or intermittent network
Week 9 Jul 19, 2026 to Jul 25, 2026	➢ Integrate i18next and allow it to dynamically read the current language from sugar-web/env. ➢ Create full locale JSON files in English, French, and Spanish for every string in the activity. ➢ Find all hardcoded strings in both the .js and .html files and replace them with i18next.t() function calls. ➢ Test the application in all three languages at runtime to ensure there are no layout issues and no missing text for any key. ➢ Implement a fallback chain so that any missing French (fr) or Spanish (es) keys fall back to English (en).
Week 10 Jul 26, 2026 to Aug 1, 2026	➢ Enhance UI/UX for every mode ➢ Polish all end game screens i.e., completion time and ranking for Number Mode, territory breakdown for Game Mode ➢ Refine the template editor UX by making dot placement and numbering intuitive enough for a child to use without instructions ➢ Ensure the full UI works correctly in both portrait and landscape tablet orientations ➢ Address all UI feedback from the mentor review at the end of Week

Week 11 Aug 2, 2026 to Aug 8, 2026	➢ Full end-to-end test of every flow : Draw Mode solo and shared, Number Mode solo and multiplayer with template sharing, Game Mode vs AI and multiplayer, template creation and Journal save/load ➢ Test in various browsers ➢ Fix all Canvas rendering bugs like dot alignment, line antialiasing, territory fill edges, color bleed between adjacent territories ➢ optimize BFS flood-fill and Ray-Casting to stay within frame budget on large grids ➢ Verify Journal save/load correctly restores all three modes including Draw Mode canvas state, Number Mode progress, and Game Mode territory maps ➢ Confirm all three locales display correctly with no errors ➢ Address all remaining mentor feedback from Week 10
Week 12 Aug 9, 2026 to Aug 15, 2026	➢ Final mentor review, address the last round of feedback and clean up remaining TODOs ➢ Cleanup and wrap up the work ➢ Complete all kinds of documentation ➢ Perform final testing and bug fixes to ensure stability ➢ Prepare the project for submission (code cleanup, final commits, and PR updates) ➢ Submit the final work product(ConnectTheDots activity) on the GSoC portal before August 24
Final Phase Aug 16, 2026 to Aug 24, 2026	➢ Every promised deliverable outlined in the proposal has been built, tested, documented and reviewed by the mentors. All feedback has been addressed the codebase is clean and merge-ready and the pull request has been submitted to the Sugarizer repository.

Following this timeline, I will be able to submit my final work product with extra features before the deadline of August 17, 2026. I have intentionally left out Replicate Mode from the scope as adding four modes to a single activity will be overcomplicating things for young learners. However, if my mentor believes it would be valuable by either integrating into Number Mode or adding as a standalone mode, I am fully capable of allocating additional hours and delivering the work without breaking the existing timeline. Therefore, even in the worst case I am confident that I will be able to complete the project before the deadline.

How many hours will you spend each week on your project?

My college summer vacations start from May 13 to July 1. In this period I can give about 15-23 hours per week and after college resumes, I can give about 14-21 hours per week. I do not have any other commitments during the summer vacation and can fully devote my time to GSoC.

How will you report progress between evaluations?

I will be active on GitHub as well as Discord I will make regular pull requests to the Sugarizer while interacting with the mentors, so anyone in the org can view my progress. Thus, my progress will always be reported thoroughly on GitHub. I will also update all the features that is adding to the project to the mentor through Discord. I will be reachable anytime through Discord, Email, or a planned video session.

Discuss your post-GSoC plans. Will you continue contributing to Sugar Labs after GSOC ends?

I am planning to continue working on the Sugarizer and MusicBlocks(SugarLab) after the program ends. During my OpenSource journey Sugarizer was the first that got my eye so I would love to be a support system to the community also I would like to explore the MusicBlocks and give contribution to there also. I am amazed by the community relations and the work carried out by this organization. I got excited by watching the admin keeps getting active daily via Github and Discord with patience just to make the community stronger. I really want to enhance my skills and put them to use to give back to the community. I aim to develop mentorship skills and the ability to guide others and try to give back to the community by mentoring and guiding others. I hope to mentor future GSoC students.

I am looking forward to contributing to SugarLabs this summer season.
Kind Regards.